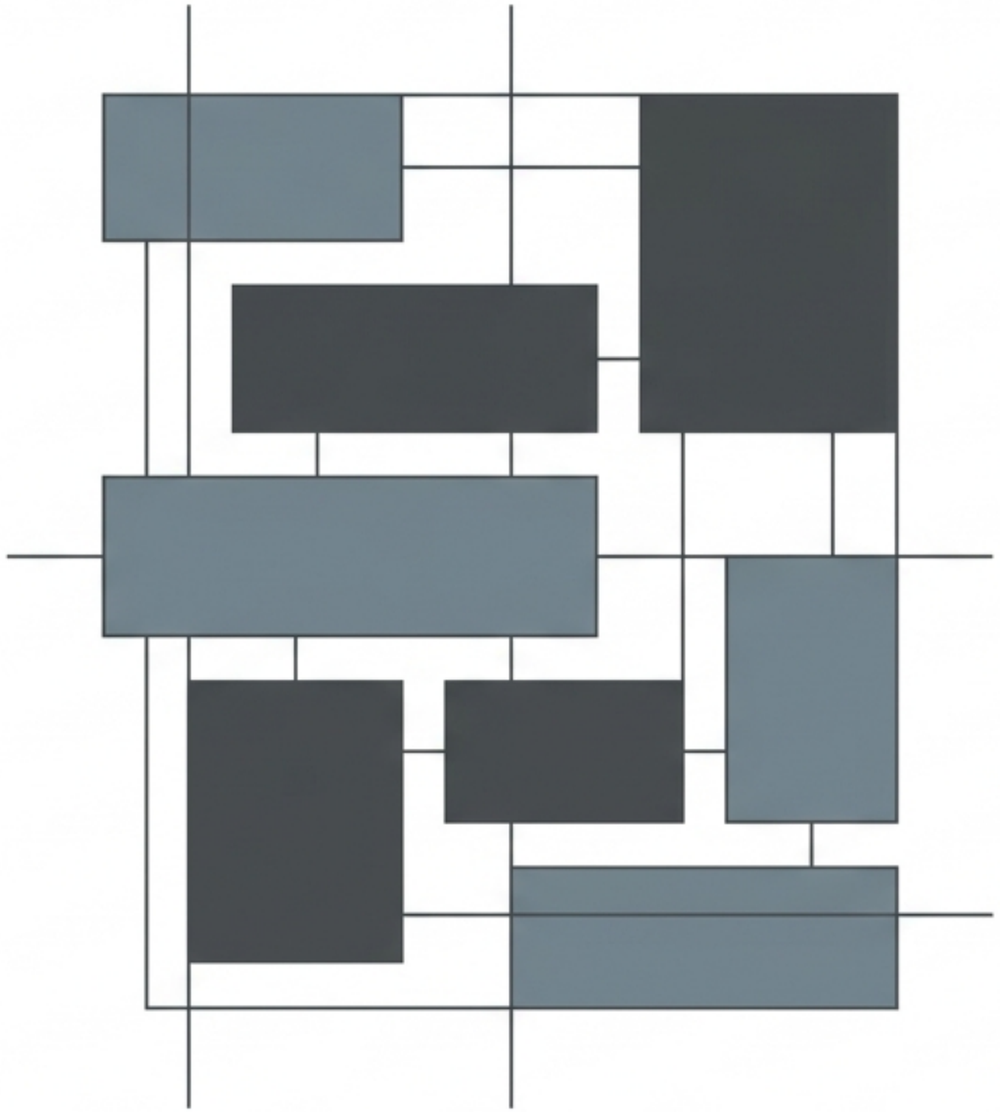


MODULE 4, LESSON 2

AI 101 for Mechanical Engineers: Self-supervised targets come from the data

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Lesson Structure

1. Concept of Self-Supervised Learning
2. Masked-Signal Reconstruction Mechanism
3. Architecture versus Learning Setup

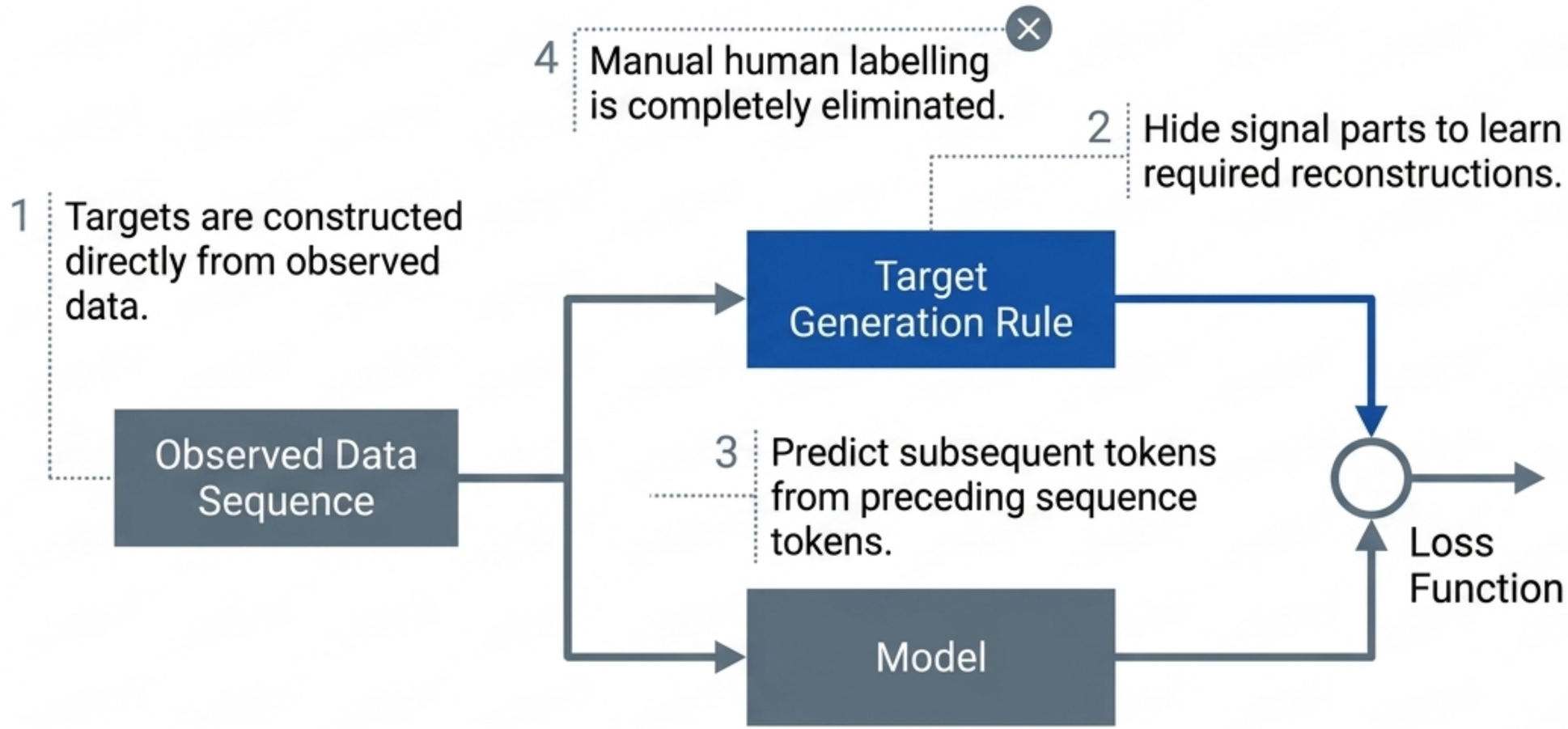
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1. Concept of Self-Supervised Learning

To define how observed data supplies internal mathematical training objectives without external human annotation.

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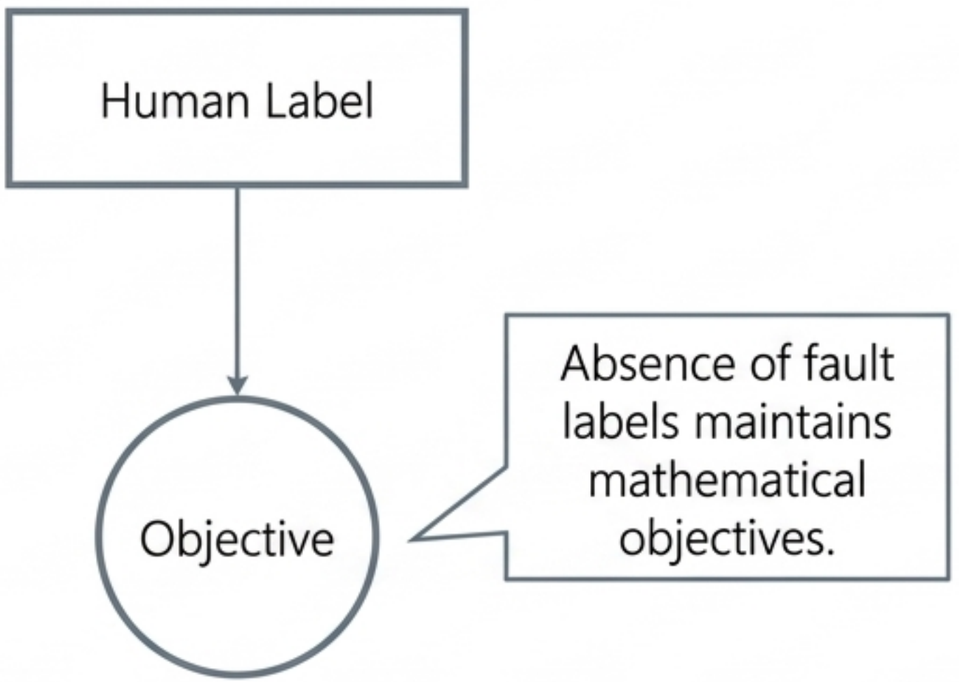
Targets Constructed from Observations



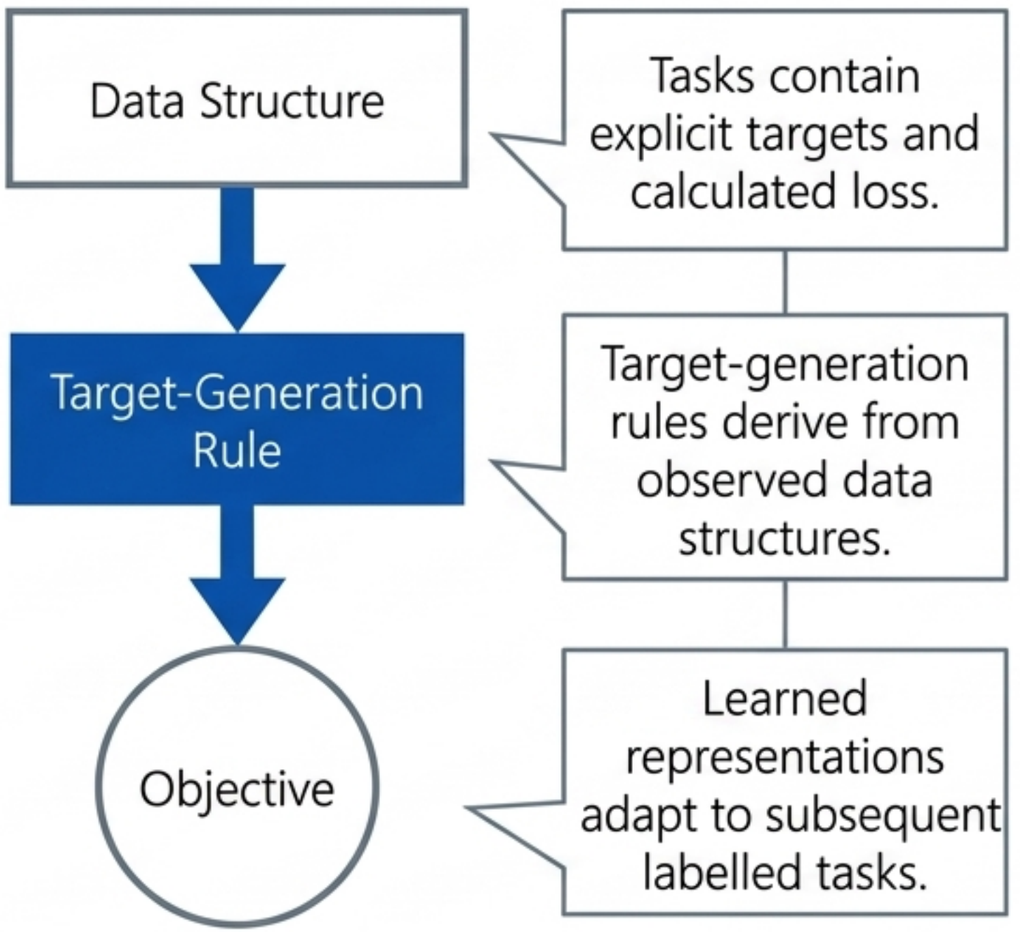
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The Self-Supervised Objective

Supervised Loss



Self-Supervised Loss



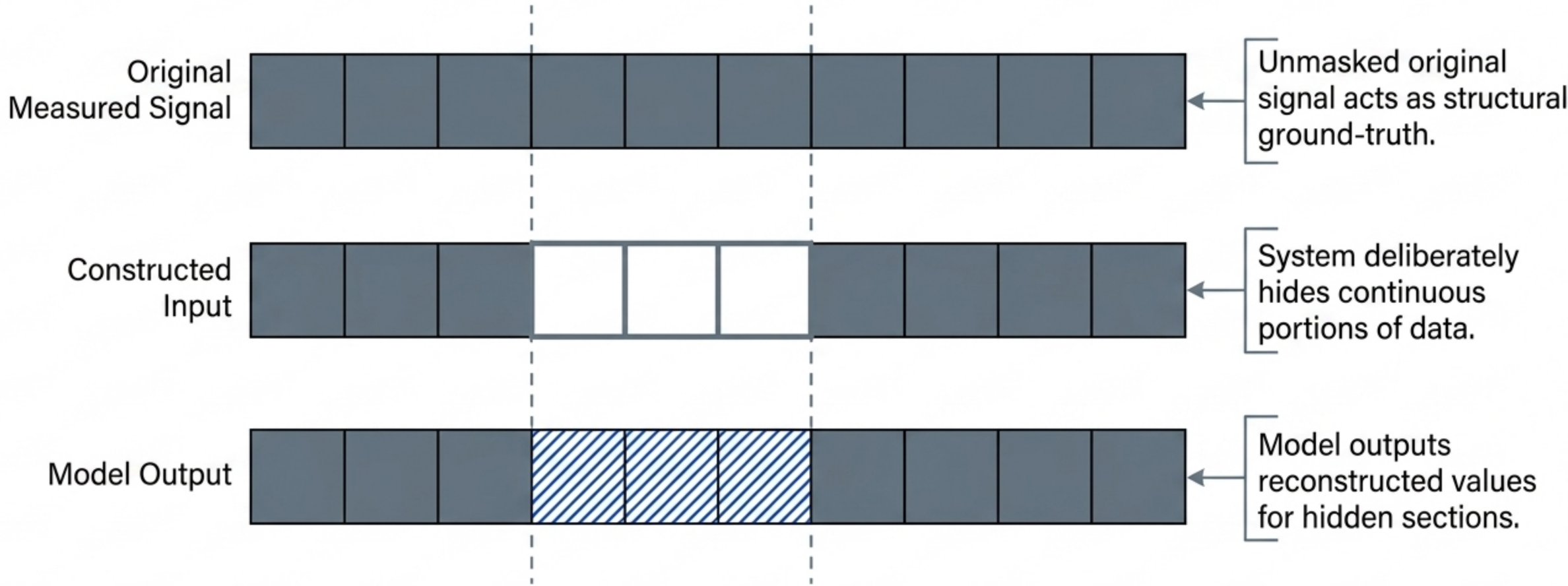
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2. Masked-Signal Reconstruction Mechanism

To demonstrate self-supervised learning through a concrete signal-processing framework.

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Masked-Signal Schematics



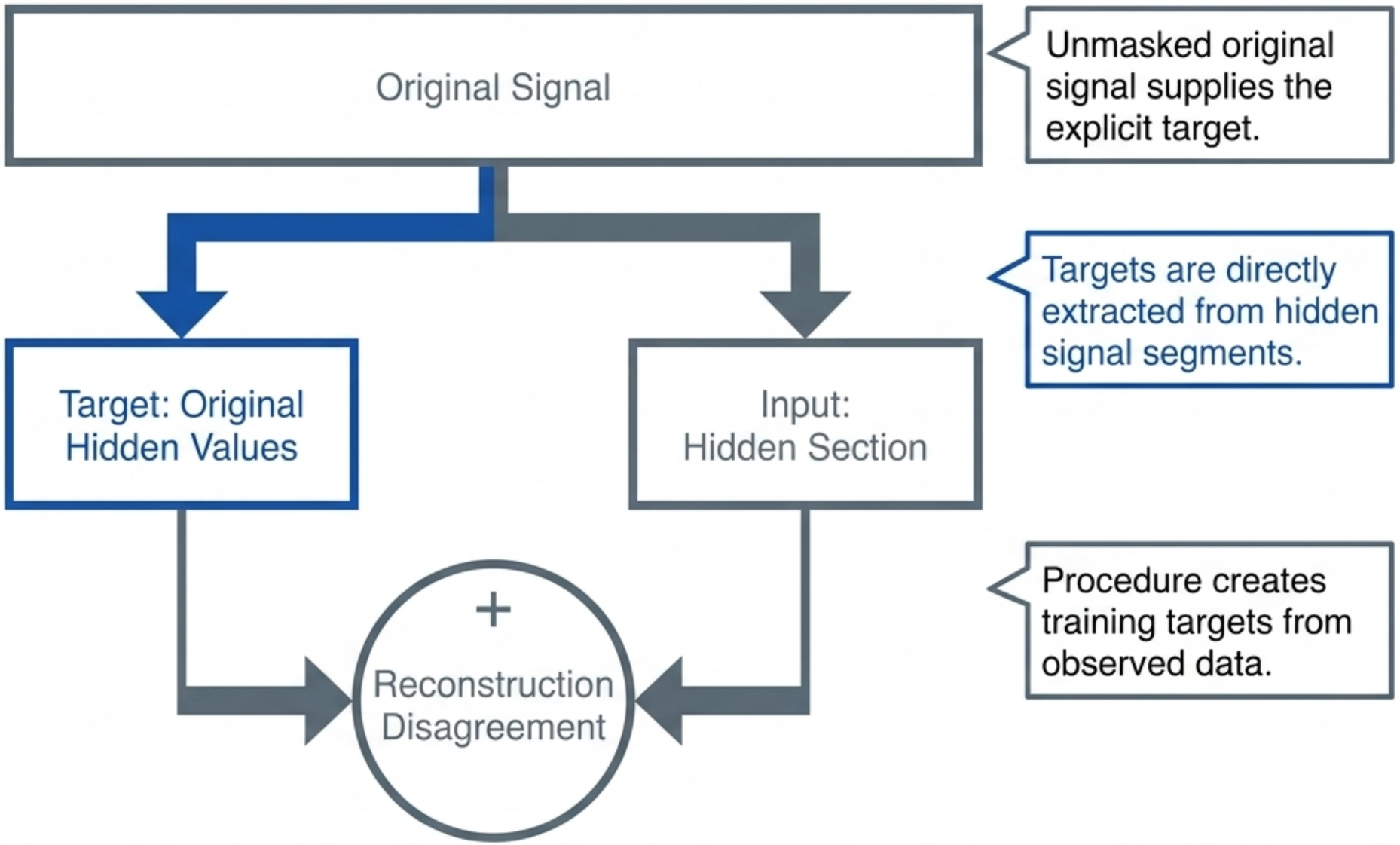
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Training Stage Matrix

Training Stage	Masked-Signal Example
Observation	An original measured signal
Constructed Input	The signal with a hidden section
Constructed Target	Original hidden values
Objective	Reconstruction disagreement

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Target Origin Derivation



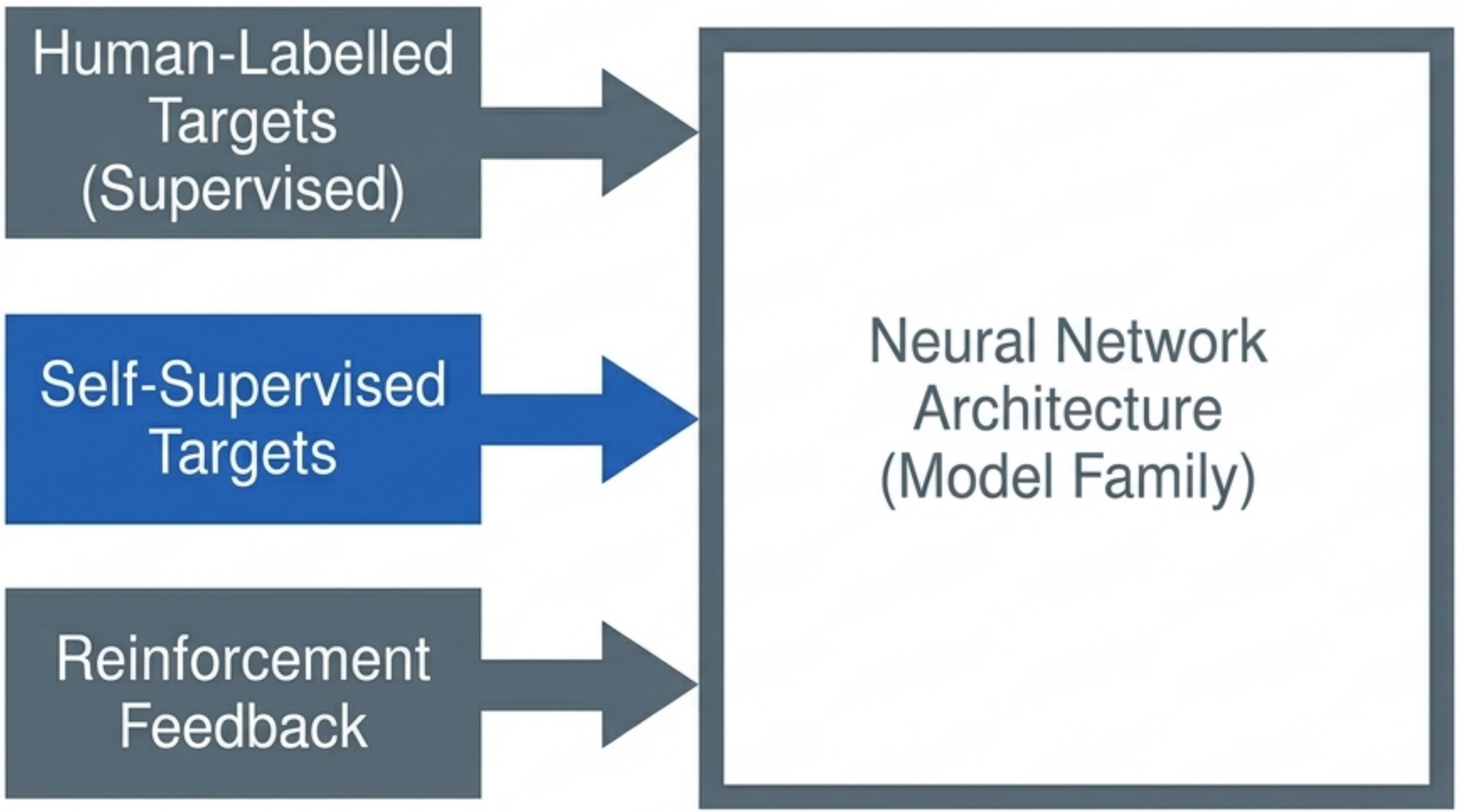
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3. Architecture versus Learning Setup

To isolate neural network model structures from their applied learning feedback mechanisms.

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Decoupling Model from Feedback



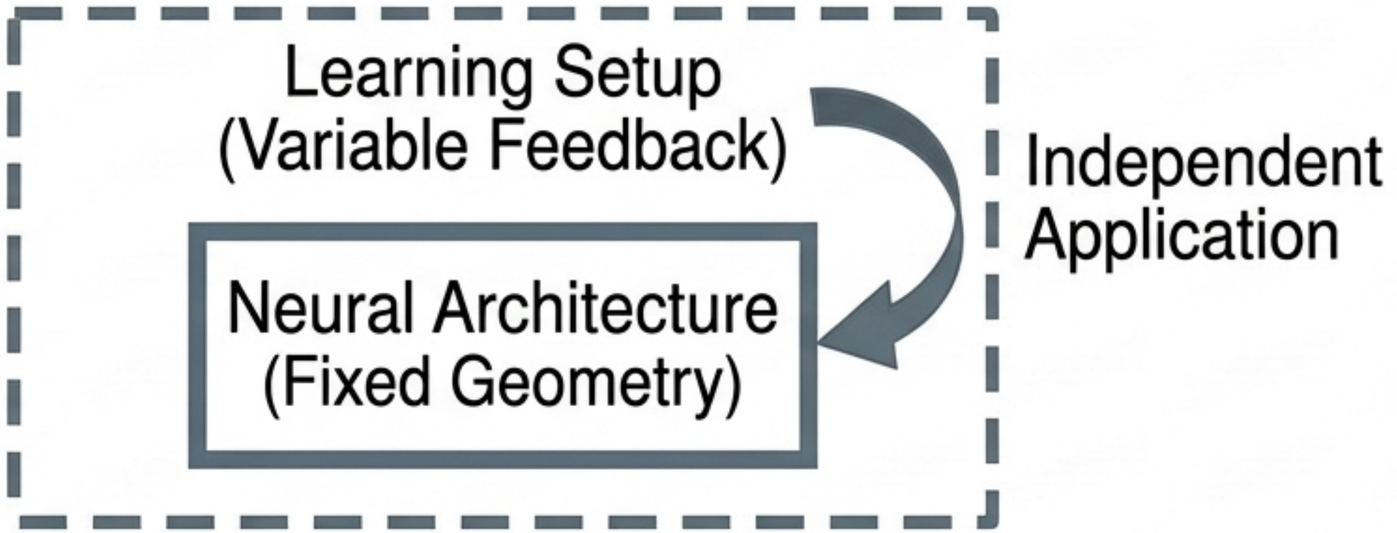
Architecture does not determine a unique learning setup.

Task, feedback, and model family operate independently.

Identical architectures support fundamentally different feedback loops.

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Setup and Feedback Principles



Reconstructing hidden portions defines self-supervised learning frameworks.

Targets are always generated from the original data.

Architecture and learning setup represent completely separate dimensions.

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Key Points

1. Self-supervised learning constructs essential training targets internally from observed data structures.

2. Masked-signal reconstruction utilizes unmasked signal segments as ground-truth targets for hidden sections.

3. Neural network architectures remain structurally independent from their applied learning feedback mechanisms.

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